

PAVAN KALYAN NEMALAPURI

B.Tech, Electronics and Communication Engineering (2023–27) | KL University
2300040115ece@gmail.com | +91-8897758735 | [linkedin.com/in/pavan-kalyan-nemalapuri](https://www.linkedin.com/in/pavan-kalyan-nemalapuri)
Portfolio: pavankalyannemalapuri-cell.github.io/portfolio



Professional Profile

Adaptable professional with a solid foundation in Java, SQL, Data Structures, Object-Oriented Programming, and DBMS, with hands-on experience in web application development and software projects. Strong problem-solving skills with a focus on developing practical, technology-driven solutions to real-world problems. Seeking an opportunity to apply technical expertise and contribute effectively in a software development role.

Technical Skills

Programming Languages: Java, SQL, C, Python (Basic).

Core Concepts: Object-Oriented Programming (Java), Data Structures and Algorithms, DBMS.

Digital Design & HDL: Digital Design, Verilog HDL, FPGA.

Tools: Xilinx Vivado, Arduino IDE, LTspice.

Education

KL University, Vijayawada — B.Tech, Electronics and Communication Engineering *2023 – 2027*
CGPA: 9.8

SR Junior College, Srikakulam — Board of Intermediate Education Andhra Pradesh *2021 – 2023*
Percentage: 98.3%

Sai Vineeth School, Palasa — Board of Secondary Education Andhra Pradesh *2020 – 2021*
Percentage: 97.5%

Projects

Automatic Railway Gate Control System

- Developed an IoT-based automatic railway gate control system using ESP8266, IR sensors, and servo motors.
- Implemented real-time train detection and automated gate control using embedded C.
- Integrated sensors and actuators to enable reliable and low-latency railway crossing operation.

The Recipe Library

- Developed a Recipe Library web application for searching and exploring recipes.
- Integrated TheMealDB API to fetch recipe details including ingredients, category, cuisine, and cooking instructions.
- Implemented API integration, event handling, and dynamic popup-based recipe display.

Smart Greenhouse System

- Developed an ESP32-based system to monitor greenhouse conditions using DHT11 and soil moisture sensors.
- Automated irrigation and temperature control using sensor-based thresholds.
- Enabled remote monitoring of environmental conditions and actuator status.

Up/Down Counter using Master-Slave JK Flip-Flops on Artix-7 FPGA

- Designed a 3-bit synchronous Up/Down counter using master-slave JK flip-flops in Verilog HDL.
- Implemented the design on an Artix-7 FPGA with a 4-digit common-anode 7-segment display.
- Added direction control and clock division, and verified functionality through simulation and hardware testing.

Achievements

- Merit Scholarship, KL University Awarded Rs. 4.8 Lakhs merit scholarship for academic excellence in Intermediate, with 50% tuition fee concession each semester.

Certifications

- Embedded System Application and IoT Programming.
- VLSI Design and Verification.